

Coalition Deception and Defensive Adaptation in a Social-Deduction Benchmark

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Abstract

How well do defenses against coordinated testimony withstand attackers trained against them? We investigate a social-deduction benchmark with structured claims, exact truth labels, and two impostors sharing a reinforcement-learning policy. An exploratory 240-job study finds higher false-ejection rates and raw vote agreement after training; this does not establish communication-driven collusion. With five crew and two impostors, descriptive multi-round coalition-victory rates are 0.967 before defender training, 0.360 afterward, and 0.851 after coalition retraining. Training optimizes victory, while innocent ejections remain frequent. Ten-generation crossplay shows renewed gains against the latest opponent, without establishing an asymptotic cycle. A hypothesis-based defender incurs substantial truthful-testimony error and is vulnerable to targeted training, which increases false ejection from 0.126 to 0.422. Ballot diagnostics support an abstention-related vulnerability under fixed recorded votes. Scripted alibi errors occur without excessive total coalition credibility weight, but scoring interventions do not identify a unique causal channel. A separate Gaussian surrogate characterizes when weight amplification is possible under explicit assumptions. Dependence penalties have mixed effects on victory and voting harm. These findings support evaluating defenses across adaptive opponents, distinct outcomes, and directly measured mechanisms.

1 Introduction

In social deduction, players combine private observations and public testimony to decide whom to eject. Agreement can reflect independent evidence or coordination within a hidden coalition. A defense that handles a scripted alibi may therefore perform differently against an adversary trained to exploit its decisions.

We study this problem in an Among Us-inspired benchmark with two policy-sharing impostors, structured claims, and explicit records of evidence, testimony, and votes. The benchmark supports both scripted attacks and learned responses. Our question is how defensive performance changes with the attacker, the training sequence, and the definition of harm.

Strategic communication connects to cheap-talk and signaling games [1], [2], trainable sender–

receiver systems [3], [4], and information aggregation with multiple senders [5], [6]. Robust aggregation, Byzantine-resilient reinforcement learning, and peer prediction offer related defenses with explicit assumptions [6], [7], [8]; geometric-median mechanisms also face strategyproofness limits [9]. Learned social-deduction agents and Among Us-style communication have already been studied [10], [11]. Adversarial communication, credit assignment, cooperation under uncertain incentives, and secret collusion provide adjacent settings [12], [13], [14], [15].

Adaptive evaluation is already an established requirement in adversarial robustness [16], as is learning approximate responses to policy populations [17]. Our contribution is the combination of adaptive-defense crossplay, separate victory and innocent-harm outcomes, and direct diagnostics of testimony scores and ballots in one structured benchmark. Claim labels describe testimony alongside voting outcomes; crossplay distinguishes opponent-specific gains from transfer. A Gaussian report model supplies a separate, analytically tractable account of weight amplification.

The results are exploratory and do not support a single ranking of defenses. Scripted benefits coexist with baseline errors and targeted vulnerability. Training objectives also differ from some evaluated harms, so improved reward does not imply uniformly better defensive decisions.

2 Methods

2.1 Benchmark and information

The benchmark contains two hidden impostors and 3, 5, or 7 crew members, denoted 3+2, 5+2, and 7+2. Roles are randomized each episode. A meeting consists of structured claims, responses, and a plurality vote among non-skip candidate ballots. Skip ballots are excluded from the tally: the unique candidate with the most votes is ejected even if most voters skip. No candidate ballots, or a tie for the highest candidate tally, produces no ejection. Ejection does not reveal roles. The incident time is public. The two impostors share a policy, reward, and a feature identifying their partner. Unless ablated, spatial policies also exchange private four-symbol messages.

The meeting-only configuration (F1–F2) generates private evidence through six random-walk steps on the Hub–Electrical, Hub–Navigation, and Hub–Medbay star map, with stay probability 0.35 and one exogenously placed incident. The spatial configuration uses the same rooms plus an Electrical–Navigation edge, eight movement steps, and two tasks per agent. An impostor can create an incident only with one crew target and no other living crew in the room; its partner may be present. Spatial experiments use either one meeting or up to eight rounds. These configurations differ in evidence generation, map, and communication, so their comparison does not isolate attack adaptation.

Players see their own evidence and the public transcript. Scripted policies use their permitted within-round positive and negative sightings; learned spatial actors receive the corresponding time-indexed private-history block, without unseen or future positions. Scripted movement

uses only available observations. Role-blind incident masks and legal no-op attempts prevent action availability from revealing the partner in hidden-identity controls. Claims receive exact true, unsupported, false, or contradictory labels from recorded states and speaker evidence. No-information is a separate action, not a truth label.

2.2 Testimony and defenses

Scripted crew choose informative supported claims and use the same suspicion rule for responses and votes. F2 compares four coalition scripts: supported testimony; a lone creator lying about its location; a creator’s false self-alibi supported by its partner; and joint scene-placement claims framing an innocent. The latter two coordinate responses and votes; without a channel, their alibi rooms need not agree. Supported testimony still allows selective disclosure and coalition-protective voting. These scripts differ in several behaviors, rather than manipulating agreement alone.

For voter v and candidate j , ordinary rules combine private evidence and speaker-specific testimony:

$$S_v(j) = \text{own}_v(j) + \text{AGG}_s[\text{signal}_v(s, j)].$$

Scene-placement claims and accusations increase suspicion; witness alibis reduce it. Personally detected lies penalize their speaker within that speaker’s row. The rules use an unweighted sum (called mean), a rescaled median or trimmed mean, or credibility-weighted sums. Credibility is -3 for a caught lie, otherwise -2 for self-contradiction, otherwise the count of compatible informative speakers. Softmax temperatures are 1.5 (soft) and 0.15 (sharp).

Dependence-aware credibility multiplies soft weights by $e^{-4\text{ex}_s}$, where ex_s measures agreement with a speaker’s closest partner beyond its average agreement. It pools claim compatibility, alibis, supportive responses, and aligned non-skip votes over three meetings. The ballot variant discounts votes and remains active with learned crew.

The hypothesis rule enumerates two-member coalitions, excluding publicly known crew, and averages constraint feasibility over honesty gates and auxiliary creator/time assignments. Each coalition speaker’s informative claims are accepted with probability one half or ignored. Infeasible hypotheses receive zero weight; positive scores are normalized. Voting requires membership score above one half, with lexicographic ties resolved by creator score, private evidence, then agent index; all-zero views abstain when legal. The solver checks emitted constraints, with a necessary-constraint relaxation for some negative transition claims. These are heuristic feasibility scores, not calibrated generative posteriors. Appendix B lists suspicion coefficients; public configurations specify the complete actions and policies.

2.3 Training and experimental design

Masked, non-recurrent actor-critic policies use PPO [18], [19] with two 128-unit hidden layers, learning rate 3×10^{-4} , discount 1, GAE parameter 0.95, clip 0.2, four phase-stratified

minibatches, and up to eight epochs. Approximate KL above 0.075 stops an update; the value coefficient and gradient-norm limit are 0.5, and the entropy coefficient declines from 0.02 to 0.005. Standard stages train for 400 updates. Each update collects complete batches of 64 episodes in F1 or 32 in spatial experiments. Unfinished trajectories are never finalized with terminal targets. Evaluation uses a copied policy and independent action randomness.

Rewards are terminal and do not directly reward lying or coordination. F1 gives the coalition +1 for an innocent ejection after a real incident and -1 otherwise. F3’s zero-sum balanced-ejection objective gives the coalition +1 for innocent ejection, 0 for no ejection, and -1 for coalition ejection after a real incident. Without an incident it gives the coalition -1 and the crew +1 regardless of the vote, including an innocent ejection. Thus incident-free voting harm is unpenalized by this objective. Single-meeting multigeneration loops retain that coalition objective but train crew with meeting-ejection rewards after a real incident: +1 for coalition ejection, -1 for innocent ejection, and 0 for no ejection or no incident. Reward controls compare these constructions. In multi-round survival experiments, the coalition wins when living impostors are at least as numerous as living crew after a meeting vote; crew win by eliminating the coalition or reaching the round cap without that condition. This reward optimizes victory, not avoidance of innocent ejections. A separate control substitutes the incident-gated innocent-ejection count for survival.

Response training proceeds $C_0 \rightarrow D_1 \rightarrow C_1$: coalition against scripted soft-credibility crew, crew against frozen C_0 , then coalition against frozen D_1 . Every stage starts fresh actor and critic weights. Multigeneration loops continue these approximate responses with matched seed offsets and budgets. They do not track a continuously updated policy. Crossplay tests frozen policies; policy distances use fixed probe observations and include initialization effects.

The corrected study comprises 240 jobs and 1,255 learner stages:

Controls remove partner identity, the private channel, or both; increase defender training to 800 or 1,600 updates where specified; and replace single-meeting or multi-round rewards. Complete designs, cohorts, and signed comparisons are versioned with the data.

Table 1. Study cohorts and evaluation budgets. Episodes and generations are repeated observations within a seed or evaluation replicate; complete job definitions and comparison expressions are versioned with the data.

Experiment	Independent units and evaluation budget
F1	Ten training seeds per population; 400 holdout episodes at genuinely untrained and final checkpoints.
F2	Ten evaluation replicates; 1,000 episodes per rule/condition/population.
F3–F4	Ten seeds per population; 1,000 final episodes for each of four pairings.
Multigeneration	Ten seeds through ten generations for 5+2 in both regimes and 3+2 multiround; five seeds through three generations for 7+2 multiround. Each stage uses 1,000 episodes; full crossplay uses 500 per cell.
Defended loop	Ten seeds and four generations.
Adaptive defenses	Ten seeds at 5+2; five at 3+2 for dependence defense and at 7+2 for hypothesis defense.
Static spatial studies and controls	Five seeds; information, defender-budget, and reward controls use matched reference seeds.

2.4 Outcomes, inclusion, and inference

For learned experiments, the independent unit is a training seed; for scripted experiments, it is an independently seeded evaluation batch. Episodes and generations are repeated observations. Every planned job passed checks of source, configuration, checkpoints, and raw records. No seed was removed or stopped for its outcome. Historical experiments, incomplete runs, and engineering smoke tests are excluded. Several implementation components changed together; historical differences cannot identify an isolated repair’s effect.

Single-meeting outcomes include false ejection, creator ejection, no ejection, and incident probability. Creator ejection and survival require an incident and are not unconditional complements. Multiround outcomes distinguish coalition victory, any innocent ejection, and mean innocent-ejection count. Whole-game records include every completed meeting, including incident-free meetings; terminal-meeting and pooled per-meeting rates remain separate. Dependence records retain the pre-ejection electorate and count historical meetings once. Conditional metrics are averaged over seeds with nonzero denominators, retaining contributing counts; undefined values are not replaced by zero.

Aligned voting is the fraction of meetings with both coalition ballots recorded in which both choose the same non-skip target; any skip counts as nonalignment. Multiround games use the terminal meeting. This raw agreement rate has no matched-independent baseline. The false-claim fraction counts claims labelled FALSE among all coalition claim actions, including non-informative claims.

Final evaluation starts at seed 1987654321 plus run seed, separately from monitoring at

987654321 plus run seed, and does not advance training randomness. Predetermined episode indices are completed without selecting faster games. F2 pairs replicate identities, while its rule/condition-specific streams do not hold evidence or complete transcripts fixed.

For 303 specified contrasts in 15 families, matched seed outcomes are subtracted before calculating mean effects, sample SDs, and pointwise 95% percentile intervals from 20,000 resamples of complete seed differences. Two-sided exact sign-flip tests preserve magnitudes and require null sign symmetry/exchangeability; pairing alone does not establish that assumption. Holm correction applies within each declared family. All contrasts, including unfavorable and nonsignificant results, are released. This post-audit specification was not prospective preregistration of the original research.

The minimum raw two-sided p-value is 0.001953125 with ten seeds and 0.0625 with five. Thus five-seed tests cannot reject at 0.05, nor can a ten-seed family with at least 26 tests pass Holm’s first threshold. Families were retained despite this limit. With five seeds, only 126 distinct bootstrap seed-count multisets are possible; their intervals are descriptive. Intervals are not simultaneous, and nonrejection does not establish equivalence. Unqualified numerical comparisons are descriptive means and sample SDs.

2.5 Mechanism diagnostics and response gaps

The descriptive mechanism study evaluates 540,000 episodes: ten replicates per population, four scripts, and nine rule/variant cells, with 500 episodes each. Cells within a condition share initial evidence; responses and votes can change. The descriptive ballot study evaluates retained hypothesis-attack checkpoints in 135 cells of 500 episodes, using ten seeds at 5+2 and five at 7+2. Removing ballots holds other recorded votes fixed and does not estimate behavioral adaptation.

For crossplay matrix M , the exploratory finite-pool gap $G_k = \max_i M_{ik} - \min_j M_{kj}$ includes later-trained policies. Simultaneous Hoeffding bands account for cell uncertainty and selection of extrema, assuming independent episode draws within each cell. This retrospective gap is neither true exploitability nor a convergence test.

3 Results

3.1 False ejection and vote agreement after training

In the meeting-only game (F1), coalition training increases false ejection from 0.045 to 0.271 at 3+2, from 0.073 to 0.2125 at 5+2, and from 0.091 to 0.194 at 7+2. All three final-minus-untrained contrasts have Holm-adjusted $p = 0.0117$ in the six-test F1 family. Raw same-target vote agreement and creator survival also increase, while the fraction of claim actions labelled FALSE decreases at every population (Figure 1). At 5+2, vote agreement rises from 0.108 to 0.316 and the false-claim fraction falls from 0.423 to 0.349. Shared policy parameters, public observations, and incentives can produce aligned votes without communication-driven collusion; no matched-independent learned-policy baseline is reported.

F1: monitored learning curves and independent holdout endpoints

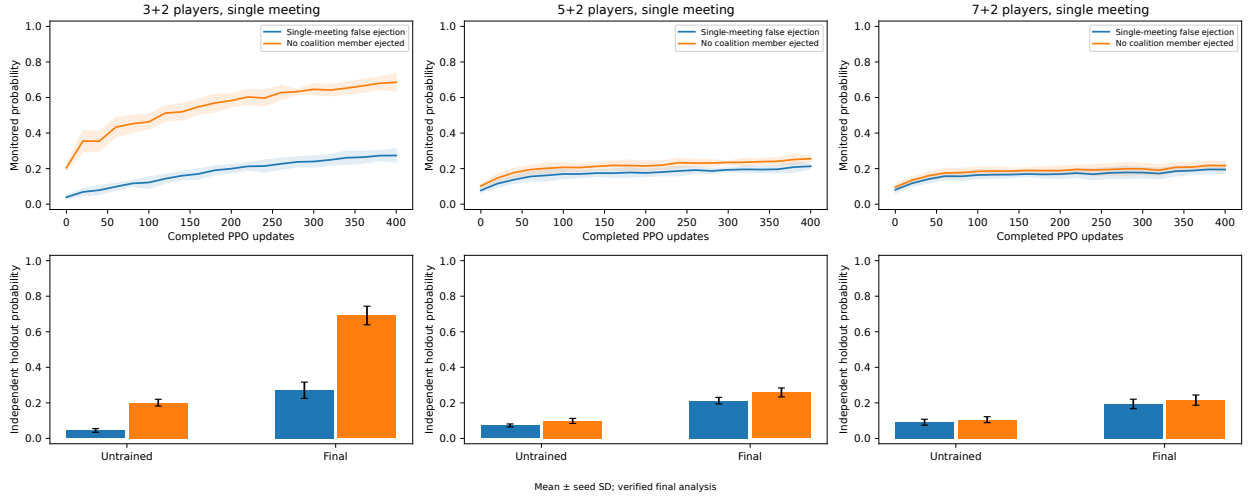


Figure 1. Coalition learning against scripted soft-credibility crew in the meeting-only game. Curves use monitored evaluations; untrained and final points use independent holdout episodes. Variability is across ten seeds. Vote agreement is raw; the false-claim denominator includes non-informative actions.

Because non-informative actions enter its denominator, the lower false-claim fraction does not establish greater truthfulness among informative claims.

3.2 Scripted performance and adaptive robustness differ

The seven-rule scripted comparison (F2) exposes an accuracy tradeoff. At 5+2, hypothesis scoring reduces false ejection relative to the mean under alibi (0.1469 versus 0.3273) and framing (0.1542 versus 0.4106), but increases it under the lone liar (0.1907 versus 0.1311). Its truthful-testimony false-ejection rate is 0.3700, whereas mean and soft credibility have zero observed errors (Figure 2). The corresponding hypothesis errors at 3+2 and 7+2 are 0.2447 and 0.3926. Here *truthful* describes supported claim contents; information selection and coalition voting remain adversarial.

Median and trimmed rules can discard useful sparse testimony, so scalar-contamination robustness does not ensure accurate votes. Hypothesis feasibility is also insufficient: a bounded replay of ten truthful-testimony errors found that the true coalition retained positive weight and its timelines satisfied the emitted constraints. Some errors arose from exact score ties resolved by agent index; another involved unanimous honest abstention. This replay identifies possible failure routes, not their prevalence. Since skips do not compete in the candidate tally, coalition ballots can decide an ejection when all honest voters abstain. The 63- and 36-test F2 families cannot reject at 0.05 with the retained ten-seed exact-test resolution.

Targeted training against the hypothesis rule (F8) exposes a further vulnerability. At 5+2, its false-ejection rate increases from 0.1255 against a soft-trained coalition to 0.4217 against a hypothesis-trained coalition: paired effect 0.2962, pointwise 95% CI [0.2521, 0.3418],

Scripted coalition and crew-rule evaluation (5+2 players, single meeting)

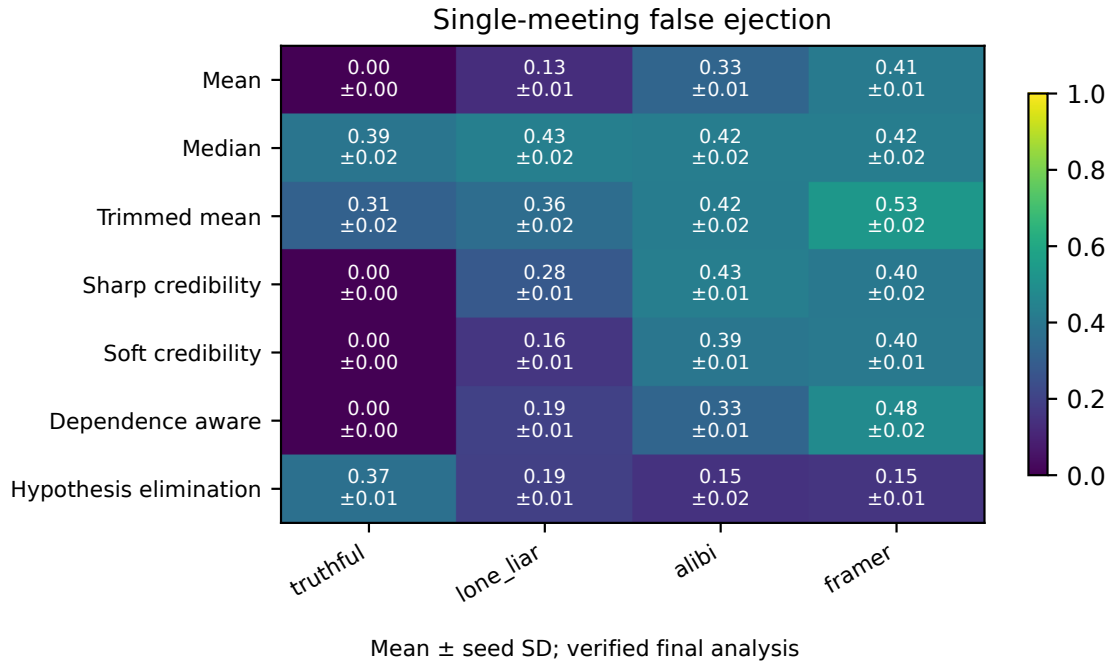


Figure 2. False-ejection rates for seven defenses against four scripted coalitions at 5+2, with ten replicates and 1,000 episodes per cell. Creator-ejection outcomes are available in the released data. These descriptive comparisons retain the full F2 multiplicity families.

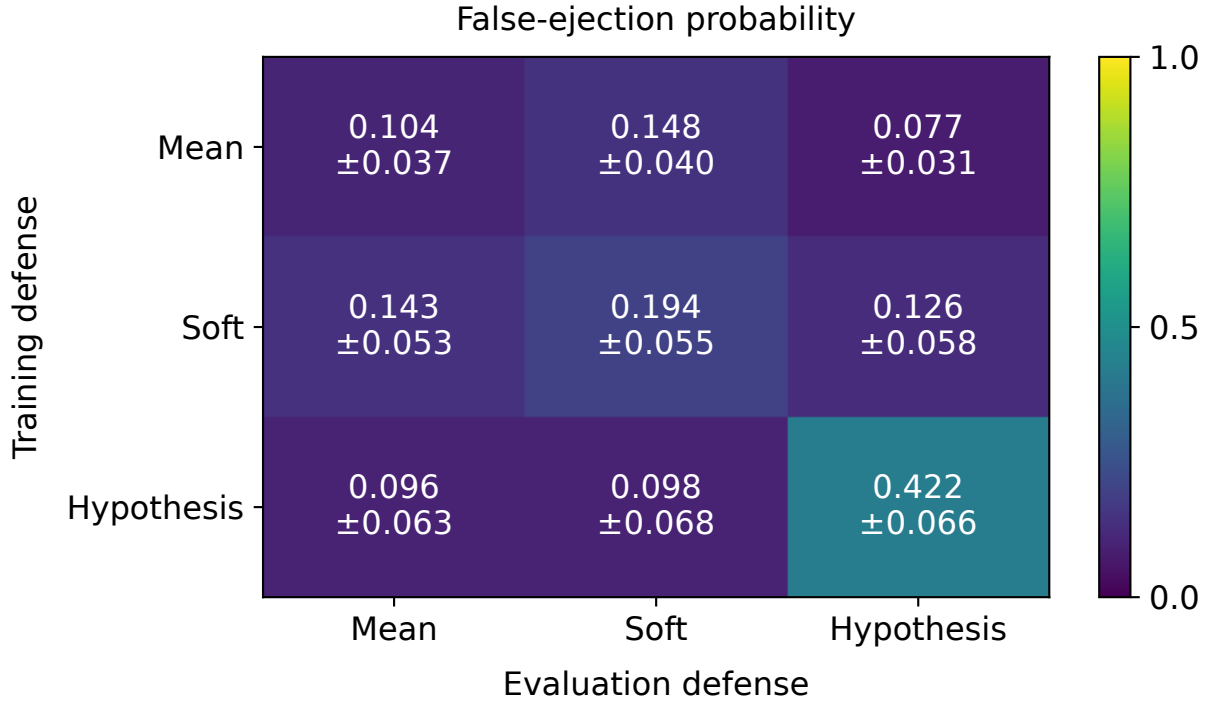


Figure 3. False-ejection rates in adaptive hypothesis crossplay at 5+2. Rows identify the coalition’s training defense and columns its evaluation defense; entries are mean $\pm$ sample SD across ten seeds with 1,000 final episodes per cell. The hypothesis-trained coalition transfers poorly to the other two rules.

Holm-adjusted $p = 0.0273$. The hypothesis-trained/hypothesis-tested pairing exceeds soft-trained/soft-tested false ejection by 0.2275 ($p = 0.0273$). Yet the hypothesis-trained attack obtains only 0.0962 against mean and 0.0976 against soft credibility (Figure 3). The attack specializes to its training defense. At 7+2 the hypothesis rate reaches 0.6036, a descriptive result based on five seeds.

An independent 67,500-episode ballot study reproduces the elevated hypothesis-attack error: 0.4240 at 5+2 and 0.6048 at 7+2. Honest voters abstain at rates 0.6608 and 0.8988. Removing coalition ballots prevents 97.26% and 100% of observed false ejections; coalition ballots alone suffice in 97.75% and 100%. Crew-only false ejection is 0.0142 and zero, respectively. Table 2 compares attack types at 5+2. These are fixed-ballot counterfactuals: they support an abstention-related route to failure, but do not predict how agents would vote under a changed rule or establish abstention as the only cause.

Table 2. Independent ballot diagnostics against the hypothesis defender at 5+2. Entries average within-seed rates over ten seeds. Necessary and sufficient condition on observed false ejection and hold recorded votes fixed; crew-only FE removes coalition ballots. Full precision, SDs, and denominator counts are released with the data.

Attack	FE	Crew skip	Necessary	Sufficient	Crew-only FE
mean	0.075	0.196	0.651	0.603	0.032
soft	0.117	0.271	0.684	0.662	0.040
hypothesis	0.424	0.661	0.973	0.978	0.014

3.3 Response training changes victory and voting harm differently

In spatial single-meeting crossplay (F3), unconditional false ejection is more frequent against the learned defender. At 5+2, rates are 0.1940 for C_0-D_0 , 0.4031 for C_0-D_1 , and 0.5385 for C_1-D_1 ; the adapted coalition obtains only 0.0759 against D_0 . The specified defender-effect contrast, C_0-D_0 minus C_0-D_1 , is negative at all three populations (adjusted $p = 0.0176$). Coalition adaptation against D_1 is not uniformly beneficial: its false-ejection effect is -0.0482 at 3+2, $+0.1354$ at 5+2, and $+0.1124$ at 7+2; the negative effect does not reject.

The incident decomposition changes its interpretation. For C_0-D_1 , total false-ejection probability equals incident-associated plus incident-free false ejection: $0.4635 = 0.0202 + 0.4433$ at 3+2, $0.4031 = 0.0903 + 0.3128$ at 5+2, and $0.4747 = 0.1552 + 0.3195$ at 7+2. Each component uses all games as its denominator; these are joint, not incident-conditional, probabilities. At 5+2, incident probability is 0.9626 against D_0 but 0.2543 against D_1 . F3 rewards the defender +1 in incident-free games even after innocent ejection. Most of its observed false ejections therefore occur outside the reward’s penalized conditions. Unconditional error alone cannot establish greater vulnerability to deceptive testimony: the comparison also changes incident opportunities and includes a reward–metric mismatch. Indeed, the joint incident-and-false-ejection rate decreases after defender training at all three populations; at 5+2 the overall increase decomposes as $0.2091 = -0.0775 + 0.2866$, with the increase confined to incident-free games. Some 3+2 seeds have no incidents against D_1 ; their conditional rates remain undefined while unconditional summaries retain all ten seeds.

Eight-round play (F4) optimizes terminal victory, while innocent ejection is an additional evaluation outcome. Descriptive victory and voting-harm rates differ (Figure 4). At 5+2, victory rates are 0.9670 (C_0-D_0), 0.3603 (C_0-D_1), and 0.8514 (C_1-D_1). The last adaptation raises victory by 0.4911 while changing the probability of any innocent ejection only from 0.8392 to 0.8467 and reducing mean innocent-ejection count by 0.4249 per game. At 7+2, victory rises by 0.5104 while mean count falls by 1.0066. Differently terminated or shorter games can therefore produce more coalition wins with fewer total innocent ejections.

Transfer again depends on population: C_1-D_0 wins 0.2182 at 5+2 and 0.0630 at 7+2, but 0.9882 at 3+2. All F4 effects are descriptive under the retained family: its minimum attainable adjusted p -value is 0.0527. Large point estimates and pointwise intervals do not override that familywise limitation.

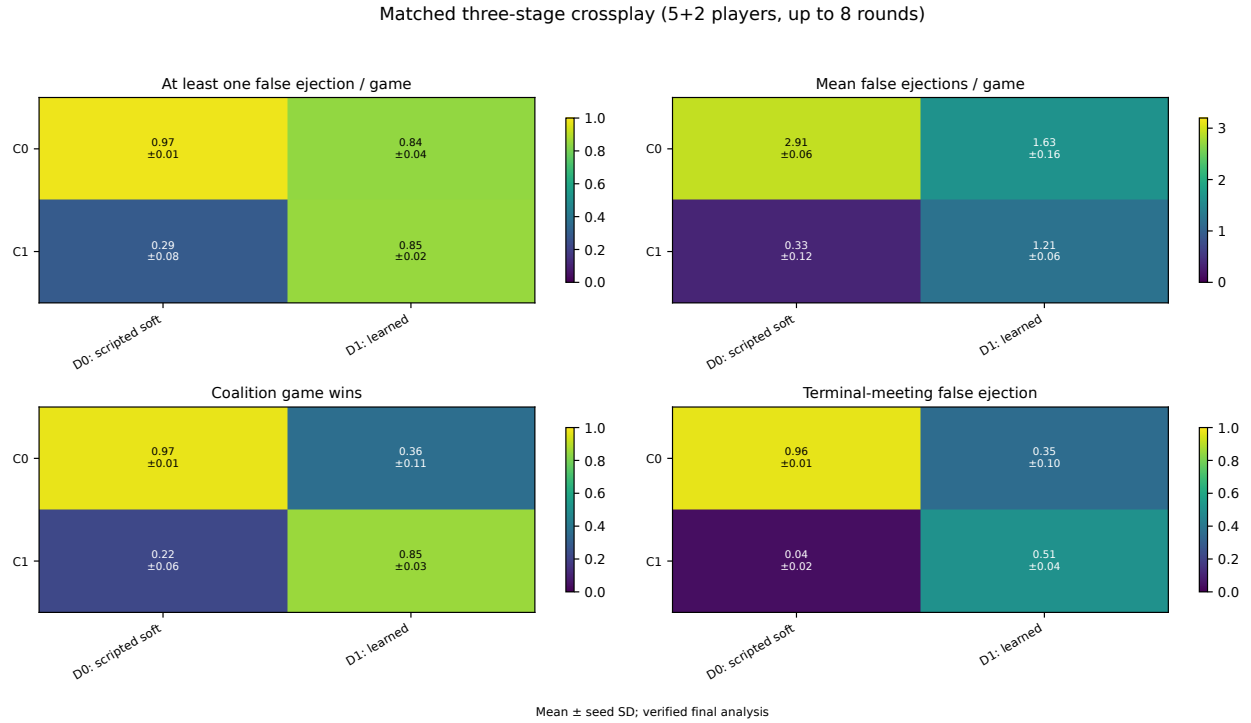


Figure 4. Eight-round crossplay at 5+2 across ten seeds. Coalition victory, any innocent ejection, mean innocent-ejection count, and terminal outcomes are distinct. Whole-game harm includes incident-free meetings. The 27-contrast F4 family has no Holm rejection at 0.05.

3.4 Repeated responses produce finite, opponent-specific gains

Ten-generation crossplay (F5) shows renewed coalition gains against the latest defender. In the separate 5+2 multi-round loop, coalition victory is 0.9658 for C_0-D_0 , 0.3524 for C_0-D_1 , and 0.8562 for C_1-D_1 . At generation ten it is 0.3068 for C_9-D_{10} and 0.7968 for $C_{10}-D_{10}$, while $C_{10}-D_0$ achieves only 0.1804 (Figure 5). These values come from the multi-generation evaluation, independently of F4.

Matrix gain averages $M_{g,g} - M_{g-1,g}$ over response generations: the outcome is false ejection for single-meeting loops and coalition victory for multi-round loops. Mean gains are 0.300 in 5+2 single-meeting play, 0.501 in 5+2 multi-round play, and 0.576 in 3+2 multi-round play. Their tests pass Holm correction in the 25-test dynamics family ($p = 0.0488$). The shorter 7+2 loop has a descriptive gain of 0.418 but only five seeds. No specified late-minus-early or parity test rejects after correction. Policy distances and finite-pool response gaps are retained in the released analysis; neither they nor the observed alternation establish convergence, nonconvergence, optimal responses, or an attracting cycle. Each stage starts from fresh weights and faces a different frozen opponent.

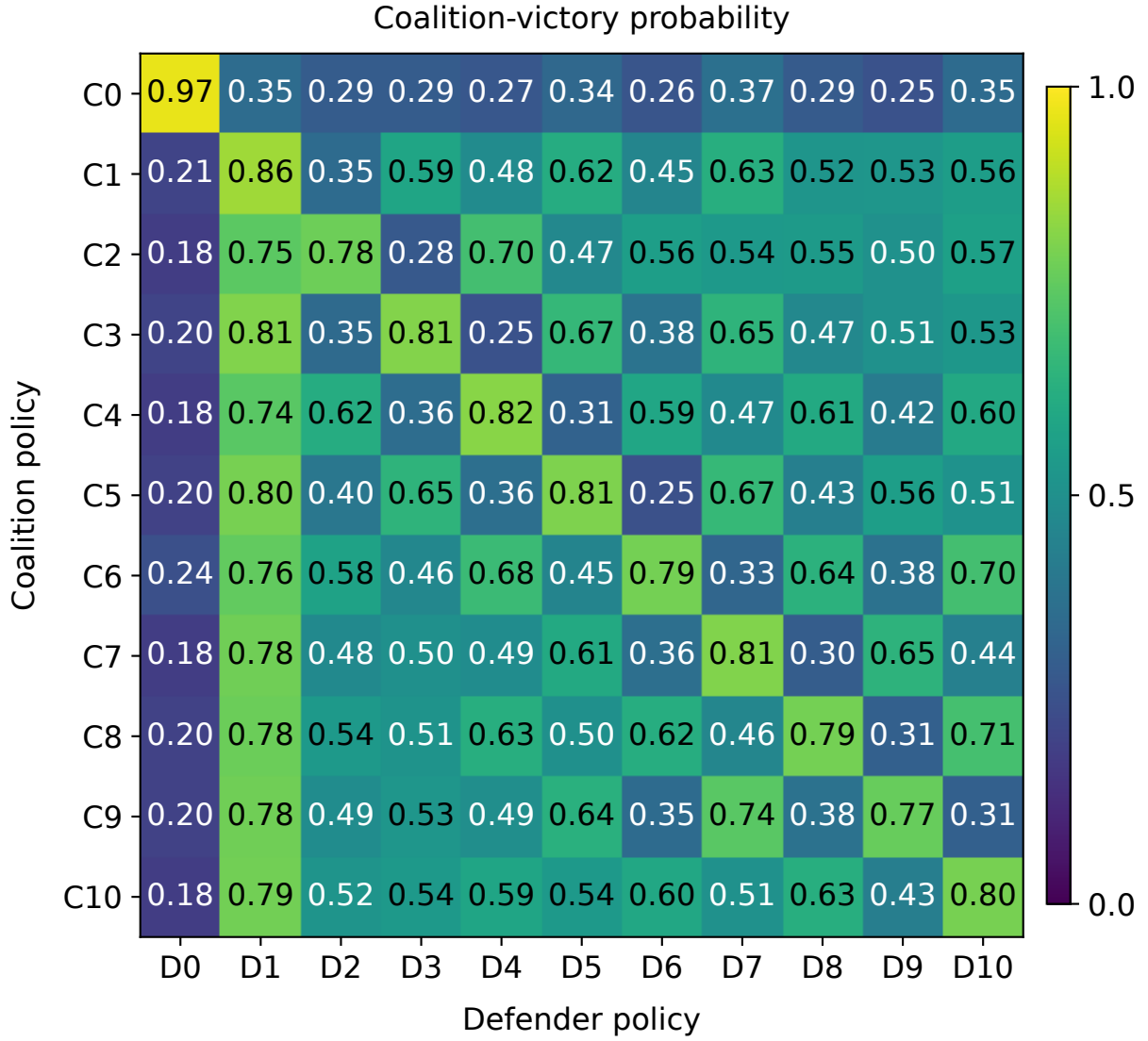


Figure 5. Coalition-victory rates in full ten-generation 5+2 crossplay, averaged across ten seeds with 500 episodes per cell. Each diagonal entry minus the entry immediately above measures coalition adaptation against a fixed defender. Poor transfer to earlier defenders distinguishes these gains from a uniformly stronger coalition.

3.5 Dependence penalties have conditional benefits

Dependence-aware defenses (F7) do not show a general adaptive-attack gain. At 5+2, training against the testimony-discount rule changes its false-ejection rate from 0.1819 to 0.1696: effect -0.0123 , CI $[-0.0516, 0.0374]$, adjusted $p = 1$. Against the combined rule and ballot discount, retraining changes 0.2044 to 0.1896: effect -0.0148 , CI $[-0.0513, 0.0215]$, adjusted $p = 1$. At 3+2, combined-defense error instead rises descriptively from 0.4132 to 0.8722; five seeds preclude a raw 0.05 rejection. None of these comparisons establishes immunity. The static spatial rule sweep likewise shows attack-dependent rankings and is distinct from F2’s exogenous-evidence comparison.

In the four-generation 5+2 loop, dependence-aware defense increases any innocent ejection by 0.042425 after coalition stages (CI $[0.034775, 0.049400]$) and by 0.104375 after defender stages ($[0.073274, 0.135025]$); both adjusted p-values are 0.0117. After coalition stages it reduces coalition victory by 0.028775 ($[-0.045625, -0.012050]$, $p = 0.0410$). Table 3 retains all six comparisons, including the nonsignificant defender-stage victory result. Smaller adaptation gains therefore coexist with greater voting harm.

Table 3. Dependence-aware versus reference loops over the same four generations and ten seeds. Values are means $\pm$ sample SD; differences are defended minus reference, and stage summaries exclude C_0 . Holm correction retains the complete six-comparison family.

Outcome	Summary	Reference	Defended	Difference	Holm p
Any FE	C stages	0.847 ± 0.009	0.890 ± 0.007	$+0.042$	0.0117
Any FE	D stages	0.658 ± 0.051	0.763 ± 0.047	$+0.104$	0.0117
Any FE	Matrix gain	0.191 ± 0.051	0.129 ± 0.046	-0.062	0.0234
Coalition win	C stages	0.817 ± 0.025	0.789 ± 0.026	-0.029	0.0410
Coalition win	D stages	0.309 ± 0.065	0.348 ± 0.054	$+0.040$	0.1387
Coalition win	Matrix gain	0.509 ± 0.086	0.445 ± 0.070	-0.064	0.0469

3.6 Information, budget, and reward controls

The five-seed controls qualify the interpretation. Removing both the private channel and partner feature gives initial-coalition single-meeting false ejection 0.2518, versus 0.1854 by default. Shared parameters, rewards, and possible role inference remain, so this does not isolate every coordination mechanism or establish that explicit communication is needed.

On matched seeds, extending defender training from 400 to 800 or 1,600 updates gives C_1 – D_1 single-meeting false ejection 0.5252, 0.5672, and 0.5160. Alternative rewards also change descriptive adaptation gains: in the four-generation single-meeting loop, matrix gains are 0.287 by default, 0.164 with balanced rewards for both sides, and 0.220 with meeting rewards for both. In the two-generation multi-round control, victory gains are 0.440 under survival and 0.333 under framing count. None of the nine outcome-ablation, eight budget, or eighteen reward comparisons rejects after Holm correction. Five-seed resolution and finite budgets prevent conclusions of equivalence or optimal training.

4 Mechanisms: score construction and correlated reports

The 540,000-episode mechanism study checks whether excessive total coalition weight accompanies errors in the tested scripted records. Under alibi attacks, soft credibility’s mean coalition weight is 0.2630 at 3+2, 0.1425 at 5+2, and 0.0976 at 7+2, below the uniform shares 0.5000, 0.3333, and 0.2500. These shares count actual reporters after the incident victim is removed. Across all 240,000 retained honest-voter records, none exceeds its uniform share at tolerance 10^{-12} . This result is specific to these scripted records.

Nevertheless, soft credibility increases alibi false ejection relative to the mean by 0.0112, 0.0680, and 0.1586 across the three populations. The score combines two relevant terms: witness alibis reduce suspicion, while a caught liar’s self-incriminating penalty sits inside that speaker’s downweighted row. Removing alibi credit, including the positive was-with subject credit governed by the same coefficient, reduces alibi errors for every tested rule and size. Moving the catch penalty into private evidence reduces soft alibi false ejection by 0.0242, 0.0912, and 0.1750. These interventions match initial evidence but allow responses and votes to change.

The benefits are conditional. Removing alibi credit increases soft framer error by 0.0876 at 5+2 and 0.1332 at 7+2; moving the catch penalty increases soft lone-liar error by 0.1540 at 3+2. Lower false ejection can also mean more no-ejection outcomes: under 3+2 alibi attacks, every tested variant has zero creator ejection and coalition-favorable rate one, meaning no coalition member is ejected, including meetings with no ejection. Under 5+2 lone liars, soft catch-outside reduces false ejection from 0.1598 to 0.1316 but increases coalition-favorable outcomes from 0.2000 to 0.2416. The narrow finding is that these alibi errors occur without excessive total coalition weight, while score modifications have condition-dependent effects. Because responses and votes change with the interventions, they do not isolate a unique causal channel or localize all failures to candidate scoring.

The Gaussian model (F6) isolates weight amplification under different assumptions. There are n honest reports with independent variance- σ^2 Gaussian noise, and $m < n$ coalition reports with bias b , marginal variance σ_c^2 , and equicorrelation ρ . Reports within distance h corroborate one another; weights are proportional to $e^{\gamma c_i}$, where c_i counts corroboration. Replacing counts by their expectations defines the surrogate coalition weight

$$W_C = \frac{f}{f + (1 - f)e^{-\gamma \Delta c}}, \quad f = \frac{m}{n + m}, \quad \Delta c = \bar{c}_C - \bar{c}_H.$$

For $\gamma > 0$, amplification $W_C > f$ occurs exactly when $\Delta c > 0$. The median instead has a uniform error bound in coalition bias when $m < n$. A long-window dependence penalty can make surrogate coalition weight non-increasing with correlation on a specified branch; Appendix A gives the assumptions, threshold, and proof. The amplification and dependence claims concern the surrogate. For example, with $n = 5$, $m = 2$, equal unit variances, $b = 20$, $h = 0.43$, $\rho = \gamma = 1$, it predicts $W_C = 0.295 > 2/7$, while 100,000 Monte Carlo trials give mean realized weight 0.277, below the uniform share.

The numerical report and sensor examples remain illustrative. For the report model at $n = 5, m = 2, h = 0.4\sigma, b = 4\sigma, \sigma_c = \sigma, \gamma = 1$, credibility MSE in units of σ^2 changes from 0.87 with independent coalition noise to 1.84 with perfectly correlated noise. The dependence-aware rule gives 0.31 in the correlated case but 1.02 in the independent case. In the twelve-sensor analogue with three coordinated compromised sensors, credibility RMSE is 0.99σ , versus 0.83σ for the mean, 0.58σ for the median, and 0.56σ for the dependence variant across twenty runs. These parameter-specific benefits do not establish a general repair. The model omits the game’s alibi credit and catch penalties, and its reporter-minority condition excludes the 2+2 electorate remaining after an incident in the initial 3+2 game.

5 Discussion and limitations

Three distinctions organize the findings. Scripted performance differs from adaptive crossplay; victory differs from voting harm; and an observed scoring intervention differs from identifying a causal mechanism. F3 leaves incident-free ejections unpenalized, while F4 optimizes victory rather than innocent-ejection avoidance. These reward–metric mismatches limit a simple interpretation of increased harm as failed resistance to deception. Raw vote agreement under shared parameters and incentives also does not establish communication-driven collusion. Adaptive crossplay and direct diagnostics characterize these outcomes without resolving those causal questions.

The scope is limited by the benchmark and learning procedure. Small maps, structured claims, and two policy-sharing impostors do not establish behavior in natural-language deliberation or independently incentivized coalitions. Non-recurrent actors retain limited cross-round memory. Fresh initialization, finite PPO budgets, and a restricted policy pool preclude claims about optimal responses or asymptotic dynamics. The hypothesis rule’s deterministic ties, constraint relaxations, and uncalibrated scores also limit what its vulnerability establishes about stronger inference-based defenses.

Statistical resolution is a material limitation, not evidence that small studies establish equivalence. The five-seed and large-family restrictions described in Methods apply even to large point estimates. The mechanism interventions preserve initial evidence rather than full transcripts; ballot removal preserves votes rather than behavior. These limits prevent unique causal explanations or universal defense rankings. Several implementation repairs changed together, so differences from historical runs cannot identify one repair’s effect. The released protocols and raw records make the reported comparisons auditable.

6 Conclusion

Training is associated with higher false-ejection rates and raw coalition vote agreement; it does not establish communication-driven collusion or greater truthfulness. Hypothesis scoring combines scripted-attack benefits with baseline error and targeted vulnerability. Finite response sequences show opponent-specific gains, while victory and innocent harm can differ under objectives that do not penalize every measured harm. These exploratory findings moti-

vate adaptive-defense crossplay, separate outcome reporting, and direct diagnostics, without establishing a universal causal mechanism or a permanent learning cycle.

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Data availability

Code, complete seed-level summaries, all planned comparisons, and diagnostic tables are available in [Coalition Deception](#). The [corrected-study-2026-09-11 release](#) contains the frozen source, checkpoints, raw episode and meeting records, environment manifests, and checksums. Appendix B describes reproduction. The complete numerical grids and extended figures remain available there, including comparisons summarized in this article.

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Appendix A. Proposition: corroboration capture under correlated testimony

Setting. Let $1 \leq m < n$, $N = n + m$, $f = m/N$, $\sigma, \sigma_c, h > 0$, $\gamma, \lambda \geq 0$, and $\rho \in [0, 1]$. Honest reports are $x_i = \theta + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Coalition reports are $y_j = \theta + b + \eta_j$, with $\eta \sim \mathcal{N}(0, \sigma_c^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top])$, independent of the honest noise.

Define $r = (x, y)$, $c_i = \#\{k \neq i : |r_i - r_k| \leq h\}$, and the realized estimator $\hat{\theta}_{\text{cred}} = \sum_i w_i r_i$, $w_i \propto e^{\gamma c_i}$. Dependence-aware weights additionally multiply by $e^{-\lambda \text{ex}_i}$, where $\text{ex}_i = \max_{k \neq i} D_{ik} - (N-1)^{-1} \sum_{k \neq i} D_{ik}$ and D_{ik} is the agreement frequency across T independent repetitions with fixed model parameters.

Exact agreement probabilities. With $s^2 = \sigma^2 + \sigma_c^2$,

$$p_{HH} = \text{erf}(h/(2\sigma)), \quad p_{CC}(\rho) = \text{erf}\left(\frac{h}{2\sigma_c \sqrt{1-\rho}}\right), \quad p_{CC}(1) = 1,$$

$$p_{HC} = \Phi((h-b)/s) - \Phi((-h-b)/s).$$

Hence $\bar{c}_H = (n-1)p_{HH} + mp_{HC}$, $\bar{c}_C = (m-1)p_{CC} + np_{HC}$ and $\Delta c = \bar{c}_C - \bar{c}_H$ is non-decreasing in ρ .

Deterministic mean-field surrogate. Replace each random corroboration count by its group expectation. The resulting coalition weight and estimator are

$$W_C = \frac{f}{f + (1-f)e^{-\gamma \Delta c}}, \quad \tilde{\theta} = (1 - W_C)\bar{x} + W_C\bar{y}.$$

This substitution defines a surrogate. In general $W_C \neq \mathbb{E}[\sum_{j \in C} w_j]$, and it does not give an exact boundary for the realized estimator. Its exact bias and mean-squared error are

$$\text{bias}(\tilde{\theta}) = W_C b, \quad \text{MSE}(\tilde{\theta}) = W_C^2 b^2 + (1 - W_C)^2 \frac{\sigma^2}{n} + W_C^2 \sigma_c^2 \left(\rho + \frac{1-\rho}{m} \right).$$

The approximation $\text{MSE}(\tilde{\theta}) \approx W_C^2 b^2$ requires the displayed variance terms to be negligible relative to $W_C^2 b^2$.

Proposition (surrogate amplification and robustness).

1. *Weight amplification.* For $\gamma > 0$, $W_C > f$ if and only if $\Delta c > 0$, equivalently

$$(m-1)p_{CC} > (n-1)p_{HH} - (n-m)p_{HC}.$$

At $\gamma = 0$, $W_C = f$. When $\Delta c > 0$, W_C increases with γ and is non-decreasing in ρ ; $W_C > 1/2$ precisely when $\gamma \Delta c > \log((1-f)/f)$. Weight amplification increases the surrogate's squared bias for $b \neq 0$; a comparison of total MSE must also include the variance terms above. At $\rho = 1$ and negligible cross-group agreement, the amplification condition reduces to $(m-1) > (n-1)p_{HH}$. Negligible cross-group agreement requires separation relative to the noise scale, for example $(|b| - h)/s \gg 1$.

2. *Median robustness and the bias reference.* Because $m < n$, the pooled median M lies between the smallest and largest honest reports for every realization, irrespective of the coalition reports. Therefore

$$\text{MSE}(M) \leq \sigma^2 \mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i|^2 \right] < \infty, \quad Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

uniformly in b and ρ . For odd N , writing $k = (N + 1)/2$ and $Z_{(j)}$ for honest order statistics gives the sharper pathwise bound $x_{(k-m)} \leq M \leq x_{(k)}$, hence $|\text{bias}(M)| \leq \sigma \mathbb{E}[Z_{(k)}]$. As n, m grow with $m/N \rightarrow f < 1/2$, the latter upper bound tends to $B_\infty = \sigma \Phi^{-1}(1/[2(1-f)])$. Thus $W_C|b| = B_\infty$ defines an asymptotic bias-comparison reference, not a finite-sample MSE break-even boundary. For $0 < W^* = B_\infty/|b| < 1$, $W_C|b| > B_\infty$ is equivalent to

$$\gamma \Delta c > \log \frac{(1-f)W^*}{f(1-W^*)}.$$

When $\Delta c > 0$ and the right-hand side is positive, division by Δc gives the positive reference gain γ^* . Finite-sample error comparisons use the actual risks or the finite-sample bound above. At $f = 1/2$, the median loses its uniform bound against arbitrarily displaced coalition reports; this does not imply that every estimator fails on every distribution.

3. *Dependence penalty.* In the long-window limit, for $m \geq 2$,

$$\bar{\text{ex}}_C = \max(p_{CC}, p_{HC}) - \frac{(m-1)p_{CC} + np_{HC}}{N-1},$$

$$\bar{\text{ex}}_H = \max(p_{HH}, p_{HC}) - \frac{(n-1)p_{HH} + mp_{HC}}{N-1}.$$

For $m = 1$, $\bar{\text{ex}}_C = 0$ and coalition correlation has no effect. The defended surrogate replaces $\gamma \Delta c$ by $\gamma \Delta c - \lambda(\bar{\text{ex}}_C - \bar{\text{ex}}_H)$. On any correlation interval where $m \geq 2$ and $p_{CC}(\rho) \geq p_{HC}$, its log-weight advantage depends on ρ through

$$\left[\gamma(m-1) - \lambda \frac{n}{N-1} \right] p_{CC}(\rho).$$

Consequently $\lambda \geq \gamma(m-1)(N-1)/n$ makes the defended surrogate weight non-increasing with correlation on that interval; a strict inequality makes it decrease where p_{CC} increases. For $5+2$, the threshold is 1.2γ . If $p_{CC} < p_{HC}$, the coefficient instead is $(m-1)[\gamma + \lambda/(N-1)]$, so the stated repair condition does not apply there. A comparison between $\rho = 0$ and $\rho = 1$ requires the first branch to hold throughout that interval.

Proof sketch. Groupwise softmax normalization gives W_C , and independence of the two groups gives the exact surrogate variance. The amplification and majority statements follow by rearranging the logistic expression. Fewer than half the reports are adversarial, so neither central rank can escape the honest range; for odd N at most m insertions can shift the median's honest rank, giving the stated order-statistic bounds. Gaussian symmetry gives the bias bound, and convergence of central order statistics gives its fixed-fraction limit. Finally, the strong law applied to each pair's agreement indicators gives the long-window matrix entries. Taking each row's actual maximum and collecting coefficients in p_{CC} yields the two dependence branches. $\square$

Appendix B. Implementation and reproduction

The original 82-file executable/configuration snapshot is identified by SHA-256 691f7687435424185064b24a9fee1832c6e7b32b9661efe20a10e35ca0c17d35. The portable package records its separate fingerprint while preserving the scientific engine and 240 job designs. Original outputs retain their original protocol identity. Complete configurations, action definitions, scripts, and analysis commands accompany the public code and release; the standard PPO and evaluation settings are given in Methods.

For scripted suspicion, private evidence scores scene sightings +3.0, sightings elsewhere −3.5, unplaced agents +1.0, and non-informative speakers +0.4. Testimony scores a caught lie +4.0 on its speaker, self-contradiction +3.0, scene placement +1.5, witness alibi −2.0, self-alibi −0.5, positive was-with claims −1.5 on the subject and −0.5 on the speaker, accusation +0.8, and contradiction +0.4. Voting requires suspicion above 0.5. Hypothesis votes use the separate feasibility and tie-breaking procedure in Methods.

The archive retains 11,885 checkpoints, including untrained, intermediate, and final weights. Checks cover source and configuration identities, finite architecture-compatible tensors, completed update budgets, matching final checkpoints, raw episode/meeting outcomes, and exact seed cohorts. All 303 contrasts in 15 families remain available, including those summarized here. Operating systems and PyTorch builds need not yield bit-identical training.

The mechanism records include 540,000 episodes and 240,000 honest-voter views. The ballot records include 135 cells, 67,500 episodes, and 317,670 honest score views. Verification reconstructs the full pre-ejection electorate, abstention, and ballot necessity/sufficiency. Two terminal-coordination cells in the 7+2 loop are undefined because a coalition voter was already lost in every game; the primary outcomes remain complete, with no seed removed or undefined rate replaced by zero. Mean catch-outside differs from its algebraically equal baseline in two 3+2 alibi episodes because a 2.22×10^{-16} score difference resolves an argmax tie; this does not support a substantive intervention effect for the mean.

This single-document edition is built from `paper_v2.md` by `build_v2.sh`, which generates standalone TeX and PDF with Pandoc and XeLaTeX. The accompanying source manifest and verifier check the retained numerical tables, figure identities, citations, and original-manuscript preservation. Full numerical grids and extended diagnostics remain in the public analysis.